Specifications for SimScape Electrical 2026

Kyle Diller

V0.1

Introduction

Since January 2026, the toolbox previously known as SimPowerSystems (SPS) is not supported anymore by Mathworks. My GitHub repository provides native Simscape Electrical (SSE) users with a library of optimized model and equivalent power systems circuits. In this document, I will explain how SimScape Electrical should be bonified to account for the new flux of power system simulation expert to the software.

SimScape Electrical was not designed by power systems engineers (as it is part of a much more general simulation tool) and therefore lacks some expertise in models, components and methods. SimPowerSystems in contrast was made with real-time and power systems in mind by Hydro-Quebec.

Nevertheless, Simscape Electrical is a powerful toolbox, only lacking some twists to make it dominant in the industry. The Simscape solver engine is a general multi-domain solver and is very similar to a nodal admittance solver like EMTP. This is a real advantage over SPS for real-time implementation as most real-time power system simulators like RTDS are based on the nodal admittance method. This nodal admittance method is really a sub-domain of the general DAE one, the former being implemented for a fixed simulation time-step and without error control. Since power system and electric drive are heavily switched models, skipping error control is really a plus. However, this comes with several tricks of the trade to make EMTP-like solver work correctly.

Now, what's wrong with SSE? Well, it's just about my past usage of SPS, it seemed so much simpler to use… This is a subjective comment, how to be sure?

What am I looking for in a library? Let's say that I hate losing time.

- Losing time searching a block
- Losing time wondering if a made errors…

The SimScape Electrical Library of 2025…

First, you should never use it from the Simulink Browser because it's a mess. Blocks in the Foundation library, then other in the Electrical one, some duplicate, some not…

Let's use the model based one, much less confusing when looking for something specific.

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>> ee_lib
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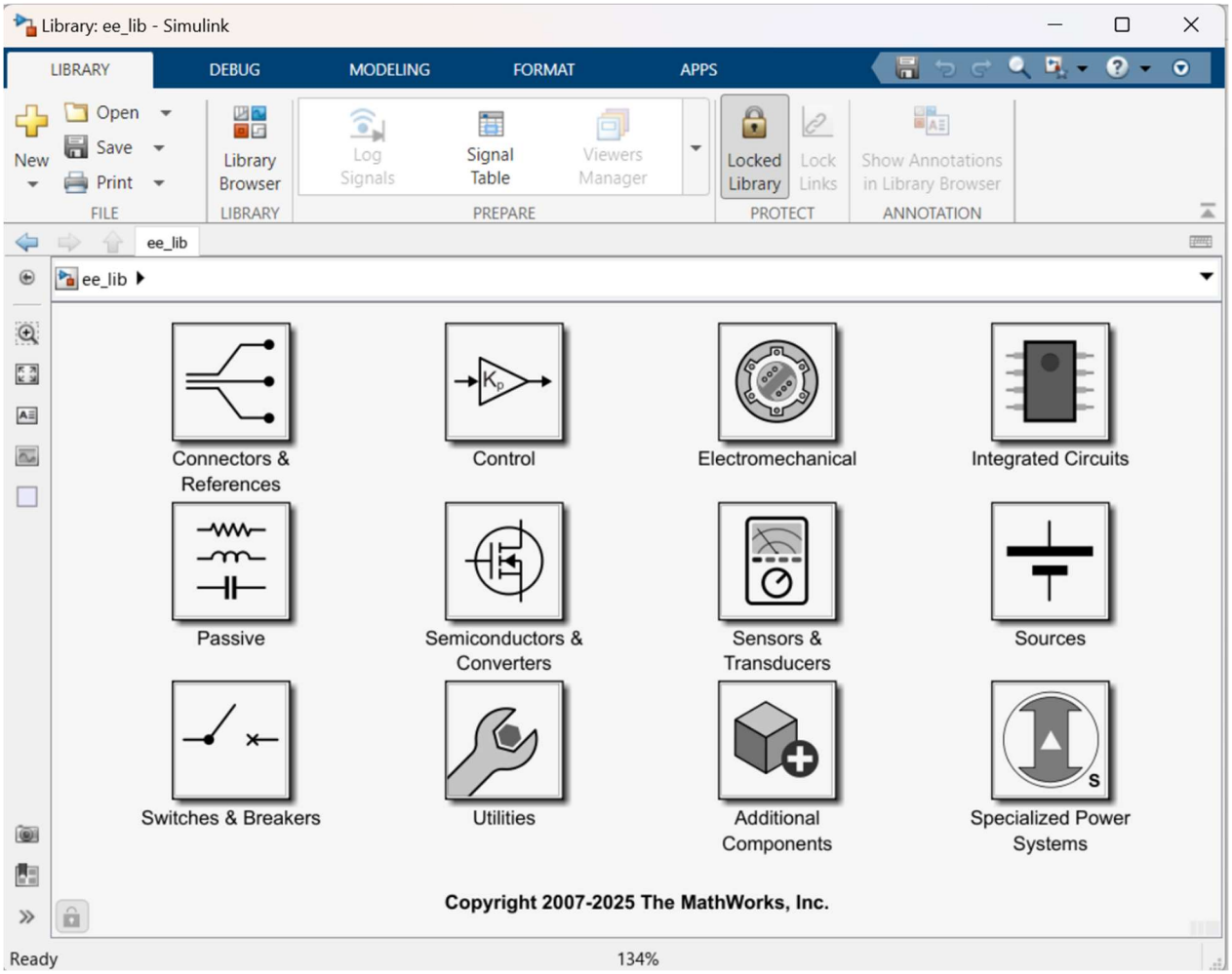
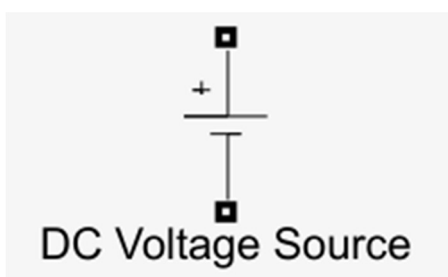


Obviously, the Specialized Power Systems is RIP now.

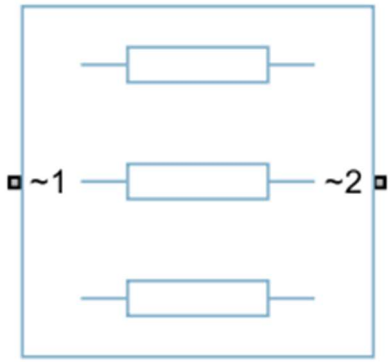
Icons

The icons in SSE are sometimes poorly designed. A DC voltage source is universally recognizable as this image:



In SSE, it's a round source, not easily discernable from the AC source with a round shape. Yes, there is a battery, but a battery is an intermediate concept.

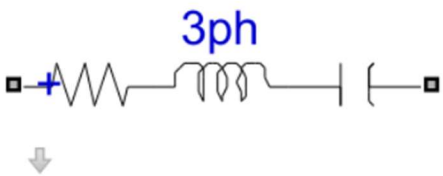
Let's take now the RLC branch of SSE



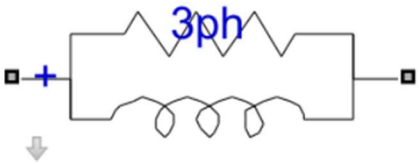
RLC (Three-Phase)

The icon stays the same whatever R, RL series, RLC parallel… When I examine and check my models, I often in doubt, they all look the same.

In MATE, I made a single block with all RLC, Series/Parallel option with changing icons.



RLC(3ph)

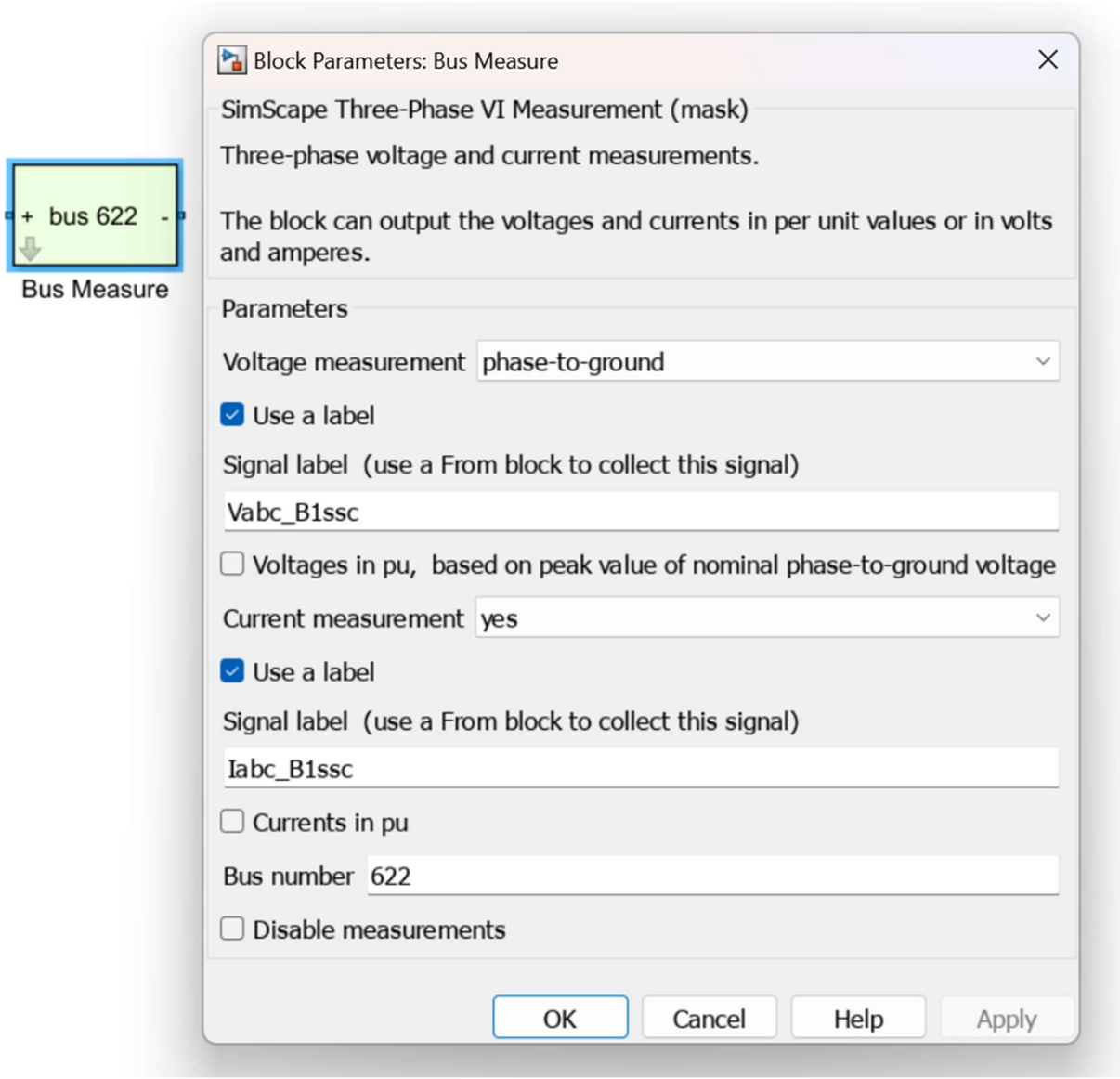


RLC(3ph)

Functionality

In SSE, there are many 3-phase sensors blocks, Current, Voltage, Voltage and Current…

In MATE, I made one to master them all.



What's nice about this block? You can read voltage and/or currents, pu/SI, bus identification, goto signal tag (to avoid lines all over the place). Disabling the measurement is also possible.

And yes, no more Simulink-Converter blocks (it's hidden inside), it outputs directly the Simulink signal of voltage and current!

Speed

Some useful SSE models are simply too slow for practical usage. One good example is the MMC-half-bridge block. The SSE detailed model seems to consider all IGBTs individually, resulting in extremely slow simulation.

In MATE, we designed a detailed model (meaning that all capacitor voltage are distinct) in an optimized way, based on [xxx]. This technique is used in RTDS and PSCAD also. Very fast, real-time capable and usable in HIL test of real HVDC systems.

Solvers

I initially found that the SimScape solver (Backward Euler and Trapezoidal) were kind of slow. That is why I designed a custom state-space solver and it runs much faster than the previous two.

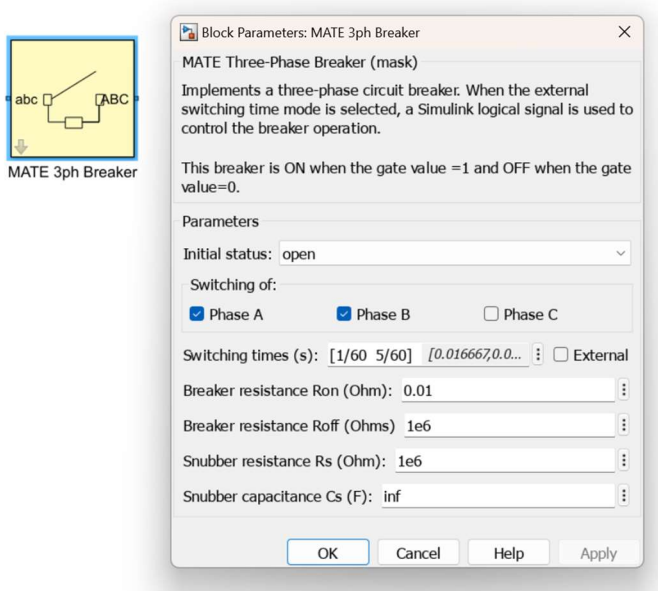
Then I tried the 'Partitioning solver' and it's very fast. Curiously, Simulink debugging tools indicates that it is using Backward Euler in this case but obviously with a better equation forming… Still, most power systems simulation tool (offline and real-time) uses the Trapezoidal rule of integration. Note than on FPGA, Backward Euler is very good because of the very small time step used.

Other basic stuff

Why does the SimScape breaker model open on logical 1? All other switching devices are all closed (ON, conducting low-impedance mode) on logical 1.

In PSCAD, RTDS, EMTP, breaker are ON with logical 1 input also.

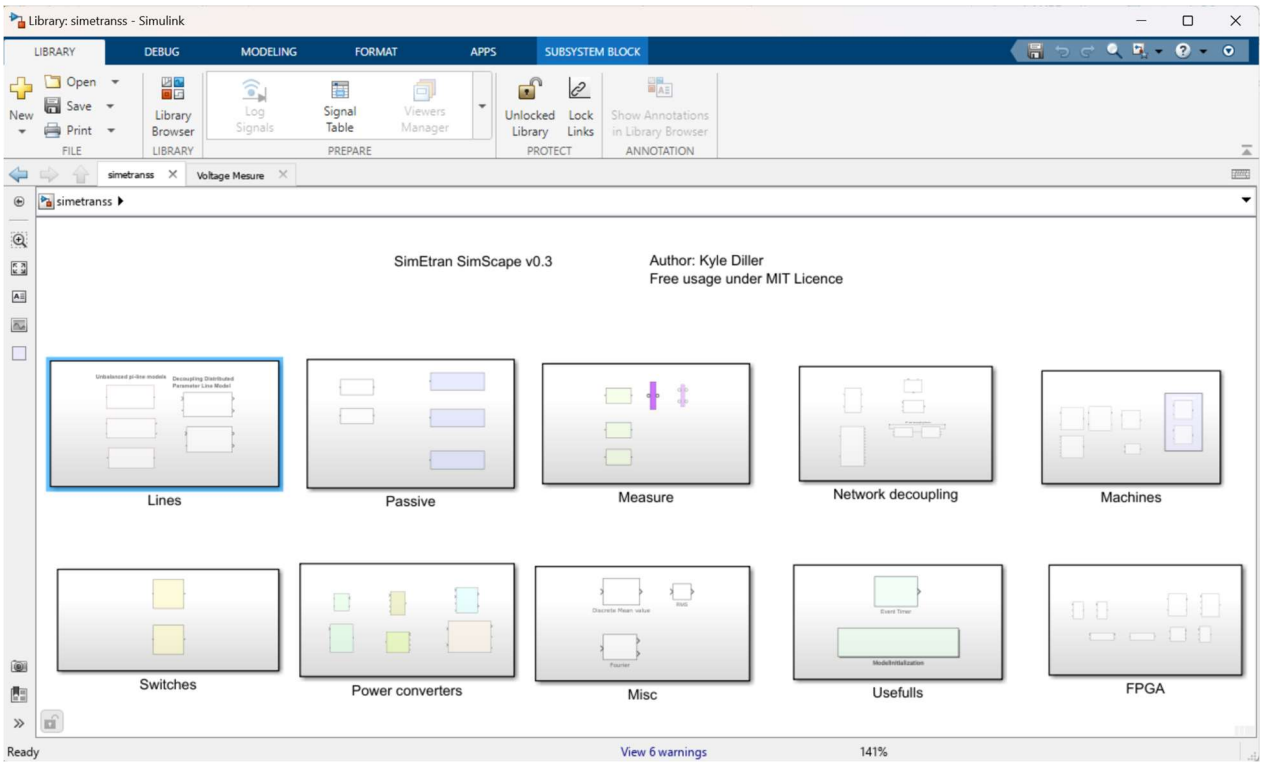
This should be corrected like in MATE.



With other useful parameters like optional RC snubber, phase selection, internal timer option…Again like PSCAD, RTDS…

The MATE library in detail.

Never forget, you can ask me to add some features in this library. Simscape Electrical is promised to a great future but big companies like Mathworks move slowly. Ask me instead, I am fast.



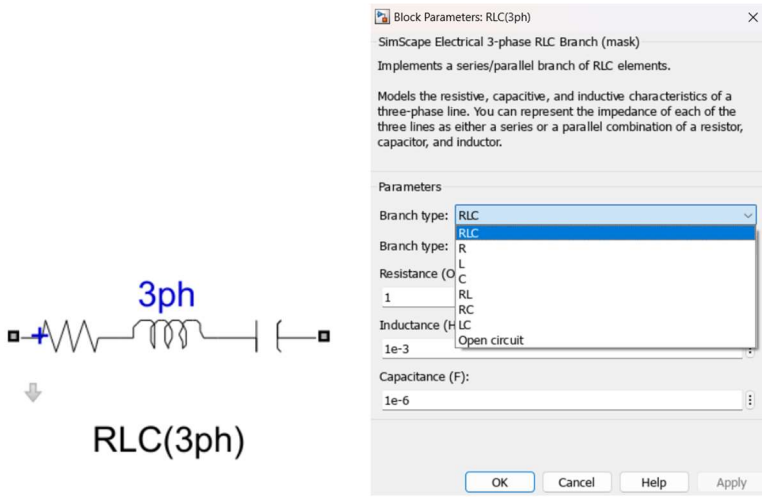
This is the SimEtrans-SimScape (SimEtrans-SS) library. Why all these blocks? I will explain.

Lines

- Unbalanced 3-phase pi-line, with either some pre-set configs or with full RLC matrix input. Sorry Mathworks, I could not find this in your library even if it is so common need.
- Distributed Parameter Line model presented in halves, for easy connection of different subsystems in Concurrent Computing mode. Even have an option to compensate for the implicit delay of the Simulink Concurrent mode.

Passive

- Unified single-phase and 3-phase RLC branches (series/parallel) and RLC Loads.

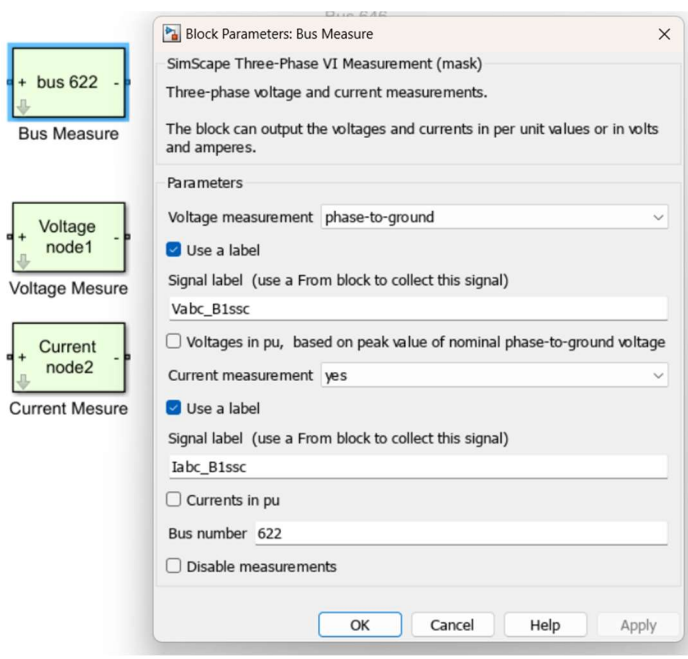


With the corresponding graphics, inspired by RIP-SPS, we make less mistakes.

Measure

Tired of all these wires and PS-Simulink converter? And with PU on top of this!

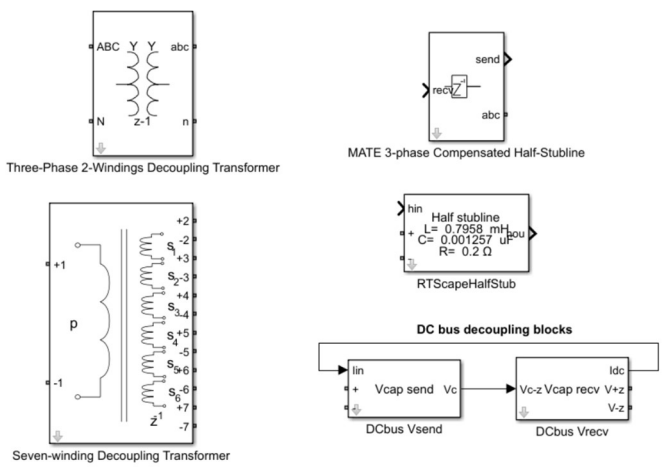
The MATE 3-phase current and voltage sensor block is for you. You can even simply label your buses with it, disabling the internal measurements (if speed is really critical). PU option is nice but most of all, your can output the reading as Goto-From signals.



Network Decoupling

This section comprises many useful block for decoupling power systems and accelerate their simulation.

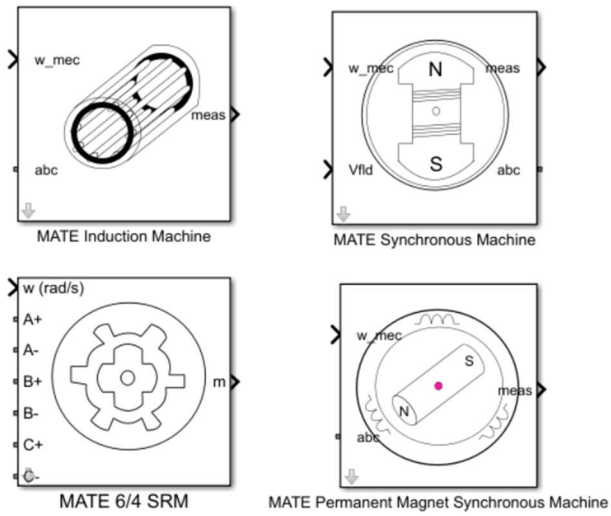
- Single-phase stubline block. A stubline is a Bergeron line model with losses (i.e. DPL) in which the delay is adjusted to 1 time-step exactly. User specify the inductance and internally the capacitance is adjusted to make the propagation delay =1 time step exactly! This is the foundation of most block of this section.
- MATE 3-phase compensated Stubline. This is a 3-phase stubline with different propagation options (to mimic even phase-to-phase capacitance) and most importantly, it is reactive-power compensated. Indeed, a stubline will add some capacitance to a network by its working principle. This reactive power can be compensated at the power frequency by adding some shunt reactance.
- 3-phase 2-winding, power transformer with stubline. This transformer model replaces the secondary leakage inductance with an equivalent stubline. A well-proven method in RTDS.
- Single-phase 7 winding stubline transformer: for a large power drive, I had to designe this decoupled model because it is not available in Simscape.



Machines

This section includes all the classic machines of Simscape Electrical but with an important twist: machine admittance, as seen by the SimScape DAE solver, is constant. This means that the models will be much faster in Simulink simulation (because the solver Jacobian will remain constant). It also means that the models are compatible with HDL Coder. The machine parameters are in SI.

- Mate Induction Machine
- MATE Synchronous Machine (PU version and Std-PU versions in dev)
- MATE PMSM : See the article about this model equations in /documentation
- MATE SRM (6/4). Other configurations can be designed on demand.



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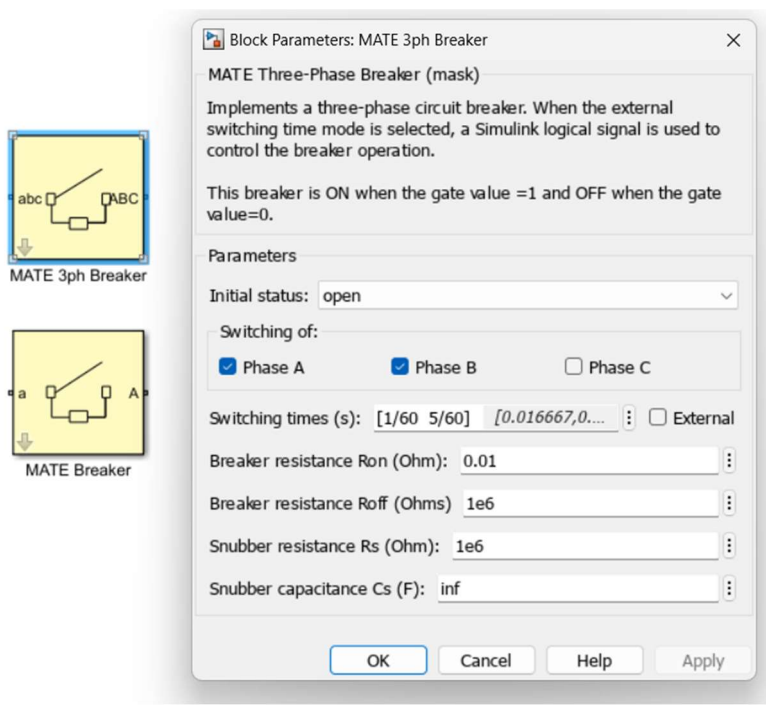
The section also contains a basic mechanical model that provides the speed input to the above machine model.

Switches

Contains breakers that have a logic consistent with ALL other switching devices, that is Conducting state (ON) when the gate=1 and Non-conducting (OFF) when the gate=0 (after zero crossing of current).

The block also contains some conveniences like internal timing of actions, RC snubber, phase selection.

In external switch control, no need for Simulink-PS converter.



Power converters

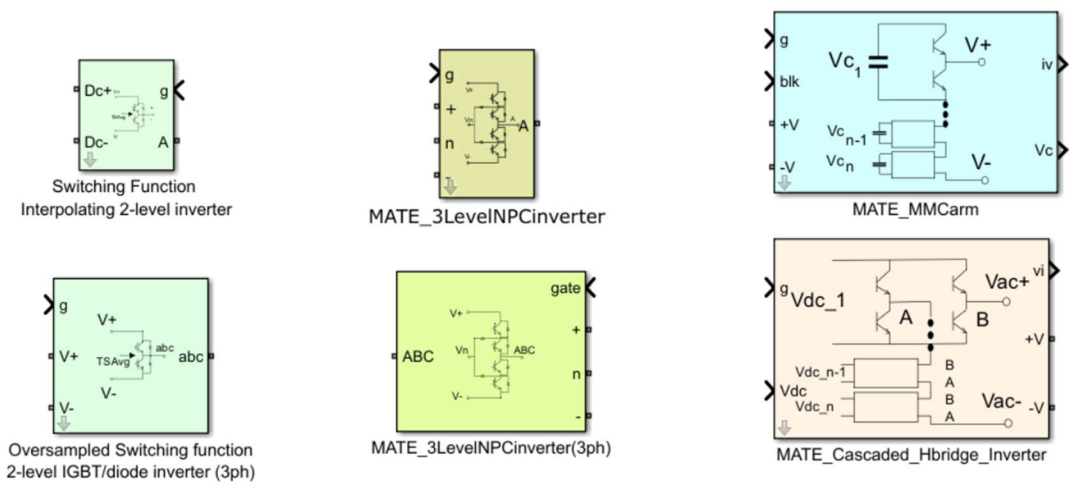
Contains many useful power converter topologies, all optimized for speed while keeping accuracy.

Switching Function Interpolating 2-level IGBT Inverter: Implements a 2-level inverter as an improved switching function. Support time-step gate averaging and high-impedance mode/rectifying modes using real Simscae switches.

Oversampled Switching function 2-level IGBT/diode inverter (3ph): same as above in 3-phase version.

MATE 3-level NPC inverters (1phase and 3-phase): Similar to the 2-level inverters and using the same techniques and benefits.

MATE MMC arms: models a complete MMC half-bridge arm, with detailed capacitors for each level. Using a very efficient C-code algorithm for speed. The C code is open in the model.



MATE Cascaded H-bridge inverter. This topology is different than the MMC in particular each stage has its own DC source. Used in a high-power converter model example with the 7-winding stubline transformer.

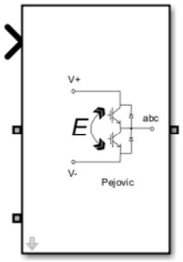
FPGA

This section contains a set of models that are particularly well-adapted to FPGA-based simulation. Their design is optimized for usage at a time step of 1-2 micro-seconds or less and they of course all have a constant admittance, making them compatible with HDL coder.

Pejovic L-C-Switch component: this block implements a Pejovic element: that is either a switch or L/C component.

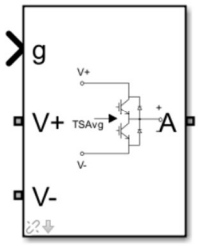
MATE 2-level inverter Pejovic: 2-level inverter made with the Pevovic element. The inverter is energy-compensated and therefore will not cause any current or voltage spikes in simulation, as opposed to Mathworks ‘Dynamic Switch’!

MATE 2-level inverter SWFHIZ: a two-level IGBT/Diode inverter implemented as a Switching Function with High impedance (Z) capability. The advantage of this inverter over the Pejovic one is that it supports oversampling and averaging of the gate inputs. This model can therefore cope with higher PWM frequency.



MATE 2-level inverter Pejovic

Pejovic L-C-switch component



MATE 2level_Inverter SWFHIZ

SimScape Electrical, HDL Coder and FPGA modeling

Other document is dedicated to FPGA simulation. Please read it and comment.